

Lab 10

Submission by: **May 16th - 4pm**

[total points: 5]

1) [0.5 points]

Show that EQ_{CFG} is co-Turing-recognizable

2) [0.3 points]

If $A \leq_m B$ and B is a regular language, does that imply that A is a regular language? Why or why not?

3) [0.3 points]

Show that if A is Turing-recognizable and $A \leq_m \overline{A}$, then A is decidable.

4) [0.5 points]

Let $T = \{\langle M \rangle \mid M \text{ is a TM that accepts } w^R \text{ whenever it accepts } w\}$. Show that T is undecidable.

5) [0.3 points]

Show that A is Turing-recognizable iff $A \leq_m A_{TM}$.

6) [0.3 points]

Show that the language $T = \{\langle M \rangle \mid M \text{ halts on empty input}\}$ is undecidable.

7) [0.3 points]

Use Rice's Theorem to prove undecidability of each of the following languages:

a. $INFINITE_{TM} = \{\langle M \rangle \mid M \text{ is a TM and } L(M) \text{ is an infinite language}\}$.

b. $\{\langle M \rangle \mid M \text{ is a TM and } 1011 \in L(M)\}$.

c. $ALL_{TM} = \{\langle M \rangle \mid M \text{ is a TM and } L(M) = \Sigma^*\}$.

8) [0.3 points]

Answer each with TRUE or FALSE

- a. $2n = O(n)$
- b. $n^2 = O(n \log^2 n)$
- c. $3^n = 2^{O(n)}$
- d. $n = o(2n)$
- e. $n^{1/10} = o(\log n)$
- f. $\frac{n^2}{1000} - 10000n = \Omega(n^2)$

9) [0.3 points]

Fill out the table described in the polynomial time algorithm for context free language recognition for string $w = baba$ and CFG G :

$$\begin{aligned} S &\rightarrow RT \\ R &\rightarrow TR \mid a \\ T &\rightarrow TR \mid b \end{aligned}$$

10) [0.3 points]

Show that P is closed under union, concatenation and complementation.

11) [0.4 points]

Let $CONNECTED = \{ \langle G \rangle \mid G \text{ is a connected undirected graph} \}$. Show that this language is in P .

HINT: In lecture 8 a TM to decide the language was provided.

12) [0.6 points]

Show that $ALL_{DFA} = \{ \langle A \rangle \mid A \text{ is a DFA and } L(A) = \Sigma^* \}$ is in P .

13) [0.6 points]

Show that $EQ_{DFA} \in P$